PRODUCT REQUIREMENTS DOCUMENT

Strategic Synthesis Engine

MediGen Corp × Tenex

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CLIENT	PREPARED BY	DATE	STATUS
MediGen Corp	**Tenex Strategy**	**March 2026**	**Draft — For Review**

1. Overview & Problem Statement

MediGen Corp's research and legal teams currently spend 60% of their working time manually reviewing a corpus of 50,000 internal documents and research reports. This document defines requirements for the Strategic Synthesis Engine (SSE) — a Retrieval-Augmented Generation (RAG) powered knowledge platform that enables staff to query this corpus in natural language and receive concise, accurate, fully-cited answers in seconds.

The Core Problem

- Research staff burn ~3 days per 5-day week locating, reading, and synthesizing documents before they can answer a single strategic question.
- Legal teams face compounded risk: manual review creates inconsistency, missed citations, and potential compliance exposure.
- At current scale (50,000 docs), the problem worsens with every new study, filing, or internal memo added to the corpus.
- The status quo is a strategic bottleneck — it delays decisions, increases cost, and creates knowledge silos across teams.

2. Goals & Success Metrics

2.1 Business Goals

- Recover at least 30% of researcher and legal staff time currently spent on manual document review within 90 days of MVP launch.
- Reduce average time-to-answer for knowledge queries from hours/days to under 3 minutes.
- Establish a defensible, auditable knowledge infrastructure that scales with MediGen's growing document corpus.
- Demonstrate measurable ROI to executive stakeholders within the first pilot period (Weeks 9–12).

2.2 Product Goals

- Deliver a query interface that returns synthesized, cited answers with end-to-end source traceability for every claim — from query to chunk to document page.
- Automatically detect and surface conflicting findings across documents, ranked by recency, so researchers never unknowingly rely on superseded evidence.
- Achieve 90%+ user satisfaction among pilot users in post-session surveys.
- Ensure the system surfaces no more than one undetected hallucination per 100 queries (verified against ground truth during pilot).
- Index 100% of the 50,000-document corpus with chunked, searchable embeddings by end of Phase 1.

2.3 Key Success Metrics

Metric	Baseline	Target (90-day)
Avg. time-to-answer a knowledge query	2–4 hours (manual)	< 3 minutes
% time spent on manual document review	60%	< 30%
Query answer accuracy (cited, verified)	N/A	>= 90%
User satisfaction score (pilot survey)	N/A	>= 4.2 / 5.0
Documents indexed in corpus	0	50,000 (100%)
Hallucination rate (undetected)	N/A	< 1 per 100 queries
Contradiction detection recall (pilot eval set)	N/A	>= 85% of known conflicts surfaced
Traceability completeness (citations per claim)	N/A	100% of claims cited to source + page
Pilot adoption rate (active weekly users)	N/A	>= 80% of pilot cohort

3. Users & Personas

The SSE serves two distinct primary personas with different workflows, risk tolerances, and definitions of a 'good answer.' Product decisions must account for both.

Dimension	Persona A: Research Scientist	Persona B: Legal Counsel
Role	PhD-level researcher; synthesizing literature to inform R&D decisions	In-house or external counsel reviewing regulatory, IP, and compliance documents
Primary need	Confidence in findings + visibility into conflicting evidence across studies, ranked by recency	Defensibility — needs exact citations, full audit trail, no ambiguity
Key pain point	Hours spent cross-referencing studies; risk of unknowingly citing superseded or contradicted findings	Risk of missed documents creating compliance gaps or litigation exposure
Definition of success	One query surfaces all relevant findings, flags contradictions, and shows which is most recent	Every claim is sourced; the system can be cited in a filing or brief
Tolerance for error	Low for uncited claims; moderate for synthesis — will verify contradictions manually	Very low — a single unsourced or unresolved conflict is unacceptable
Technical fluency	High — comfortable with filters, confidence scores, Boolean logic	Moderate — prefers clean UI, clear provenance
Usage pattern	Deep, exploratory queries; contradiction review; multiple follow-ups	Precise, targeted queries; single-session answers with export

Secondary Users

- Executive / Strategy Team — occasional high-level queries to brief leadership; need executive summaries over deep dives.
- IT / Platform Admin — manages ingestion pipeline, user access, and system health. Not a query user but a critical operator persona.

4. MVP Scope & Feature Prioritization

4.1 In Scope — MVP

Feature	Description
Document Ingestion Pipeline	Automated ETL process to parse, chunk, embed, and index all 50,000 documents into a vector database. Supports PDF, DOCX, and TXT formats. Document date metadata is preserved and indexed for recency weighting.
Natural Language Query Interface	Web-based UI where users type questions in plain English and receive synthesized answers. No Boolean syntax required.

Feature	Description
Cited Answer Generation with Full Traceability	Every claim in a generated answer is backed by an inline citation linking directly to the source document, exact page number, and highlighted passage. Uncited claims are structurally impossible — the system prompt enforces citation-only generation.
Confidence Scoring	Each answer displays a confidence score (High / Medium / Low) derived from cosine similarity of top retrieved chunks. Low-confidence answers trigger a human-review flag automatically.
Contradiction Detection Engine	When retrieved chunks contain semantically conflicting findings, the system automatically surfaces a conflict alert inline within the answer. Conflicting sources are ranked by document recency. The most recent finding is highlighted as the current best evidence, with links to all conflicting documents.
Recency Weighting in Retrieval	Retrieval scoring applies a recency multiplier so that newer documents rank higher when semantic similarity is equivalent. Document publication/creation date is a first-class metadata field used in ranking.
Source Preview Panel	Side panel displaying the top retrieved source documents with highlighted relevant passages for user verification. For contradiction alerts, both conflicting passages are shown side-by-side.
Full Trace View	Users can expand any answer to see the complete evidence chain: query → retrieved chunks (with similarity scores) → LLM synthesis → cited output. Every step is inspectable.
Query History	Users can view and re-run their previous queries within the same session and across sessions (last 30 days).
Basic Role-Based Access	Two access tiers: Standard (researchers) and Legal (adds audit log). Admin tier for IT.
Pilot Analytics Dashboard	Internal dashboard tracking query volume, avg. response time, confidence score distribution, contradiction detection frequency, and user satisfaction ratings.

4.2 Explicitly Out of Scope — MVP

Feature	Rationale for Deferral
Custom model fine-tuning	High cost, long lead time, and RAG achieves target accuracy without fine-tuning in MVP window. Revisit in Phase 2.
Multi-modal support (images, tables, charts)	Significant engineering lift. Text-based corpus covers 85%+ of query needs. Defer to Phase 2.
External database integration (PubMed, FDA)	Introduces data governance complexity and latency. Internal corpus is the defined scope.
Slack / Teams / Email integration	Valuable but non-critical for MVP. Users can access via web app. Phase 2 delivery.
Automated report generation	Out of scope for query engine MVP. Can be layered as a feature post-pilot.
Document upload by end users	IT-managed ingestion is sufficient for MVP. Self-serve upload adds access control risk.
Real-time document sync	Batch ingestion (daily) is sufficient for MVP. Real-time sync is a Phase 2 optimization.

5. Features & Acceptance Criteria

F-01: Document Ingestion & Indexing

Criterion	Detail
AC-01	System ingests PDF, DOCX, and TXT files from the designated document store without manual intervention.
AC-02	Documents are chunked into segments of 512 tokens with 50-token overlap to preserve context across chunk boundaries.
AC-03	Each chunk is embedded using a production-grade embedding model and stored in a vector database with metadata (source file, page, date, department tag).
AC-04	Full ingestion of 50,000 documents completes within 72 hours of pipeline activation.
AC-05	Ingestion failures are logged with error codes; failed documents are flagged for manual review without blocking the pipeline.
AC-06	Daily incremental ingestion runs automatically for any new documents added to the store.

F-02: Natural Language Query Interface

Criterion	Detail
AC-01	User enters a plain-text query of up to 500 characters and receives a response within 10 seconds for 95% of queries.
AC-02	The interface provides a query suggestion / autocomplete for common query patterns to reduce time-to-first-query.
AC-03	Users can apply filters for document date range, department, and document type before or after querying.
AC-04	The interface is accessible via modern browsers (Chrome, Firefox, Safari, Edge) with no installation required.
AC-05	The system gracefully handles ambiguous queries by returning its best answer alongside a prompt: 'Did you mean to ask about X or Y?'

F-03: Cited Answer Generation & Full Traceability

Criterion	Detail
AC-01	Every factual claim in a generated answer is followed by an inline citation in the format [Doc Title, p. X]. The LLM system prompt structurally enforces this — the model cannot generate an uncited factual claim.
AC-02	Citations are hyperlinked — clicking opens the source document at the cited page/passage in the Source Preview Panel, with the relevant sentence highlighted.
AC-03	Answers that cannot be substantiated by retrieved documents display a clear disclaimer:

Criterion	Detail
	'Insufficient source coverage — verify manually.' The system does not synthesize beyond what the corpus supports.
AC-04	Answer length is calibrated to query complexity: simple factual queries return 1–3 sentences; synthesis queries return structured summaries with subsections.
AC-05	Users can click 'Show all sources' to expand the full ranked list of retrieved documents (up to 10) with cosine similarity scores and document dates.
AC-06	Users can click 'Show Full Trace' on any answer to inspect the complete evidence chain: query input → top-N retrieved chunks (with scores) → LLM context window → synthesized output. This is the full audit trail of how an answer was constructed.

F-04: Confidence Scoring & Human-Review Flag

Criterion	Detail
AC-01	Each answer displays a confidence badge: High (top retrieved docs score > 0.85 cosine similarity), Medium (0.65–0.85), or Low (< 0.65).
AC-02	Low-confidence answers display a yellow warning banner: 'This answer has limited source support. Manual review recommended.'
AC-03	Users can manually flag any answer for human expert review, regardless of confidence score, with a one-click 'Flag for Review' button.
AC-04	Flagged answers are logged in the admin dashboard with query text, answer, and flagging user for follow-up.

F-05: Contradiction Detection Engine

This is a core research-facing feature. When the retrieval layer surfaces chunks with semantically opposing claims, the system does not silently average them into a misleading synthesis. It explicitly flags the conflict, surfaces all positions, and identifies the most recent evidence.

Criterion	Detail
AC-01	When 2 or more retrieved chunks contain findings that contradict each other (determined by LLM semantic comparison of claim pairs), the system inserts an inline conflict alert before the synthesized answer.
AC-02	The conflict alert displays the number of conflicting sources and a plain-English summary of each position. Example output: 'Evidence conflict detected across 3 studies. Study A (2019) reports X. Study B (2023) reports Y. Most recent evidence supports Y. [Link to Study B, p.4]'
AC-03	Conflicting sources are sorted by document date descending — the most recent source is presented first and explicitly labeled 'Most Recent Evidence.'
AC-04	The conflict alert includes direct hyperlinks to each conflicting document at the specific cited passage, not just to the document root.
AC-05	Users can click 'Expand Conflict Detail' to see the full text of each conflicting passage side-by-side in the Source Preview Panel.
AC-06	If contradictions exist but confidence in the conflict detection is low (similarity scores between claims are borderline), the system flags it as a 'Possible conflict — verify

Criterion	Detail
	manually' rather than a definitive conflict alert.
AC-07	Contradiction detection runs as part of the retrieval-synthesis pipeline — it is not a post-processing step. Latency budget for contradiction analysis is < 2 additional seconds on top of base query latency.
AC-08	The pilot analytics dashboard tracks contradiction detection frequency by topic area, surfacing patterns in where the corpus has unresolved scientific disagreements.

F-06: Recency Weighting in Retrieval

Semantic similarity alone is insufficient for a research corpus where newer findings supersede older ones. Recency weighting ensures that when two chunks are equally relevant, the more recent document ranks higher.

Criterion	Detail
AC-01	Document publication date (or last-modified date if publication date is unavailable) is extracted during ingestion and stored as a metadata field on every chunk.
AC-02	The retrieval scoring function combines cosine similarity with a recency multiplier using a configurable decay function (default: linear decay over 5 years, configurable by admin).
AC-03	The recency multiplier can be toggled on or off per query. Users can switch to 'Pure Relevance Mode' (semantic similarity only) if they explicitly want older documents prioritized.
AC-04	Every source displayed in the answer or Source Preview Panel shows the document date prominently, allowing users to contextualize the evidence age at a glance.
AC-05	The recency decay parameters (half-life, decay curve) are configurable by admins in the platform settings without requiring a code deployment.

F-07: Audit Trail (Legal Tier)

Criterion	Detail
AC-01	All queries, answers, and source citations for Legal-tier users are logged with timestamp, user ID, and session ID in an immutable audit log.
AC-02	Legal users can export any answer — including full citations, contradiction alerts, source metadata, and the full trace view — as a PDF report with one click.
AC-03	Audit logs are retained for a minimum of 7 years in compliance with standard legal document retention requirements.
AC-04	Audit log access is restricted to Legal-tier users and admins. Standard Research users cannot view legal query logs.

6. User Experience & Key Flows

6.1 Core Query Flow

Flow: Research Scientist — Literature Synthesis with Contradiction Detection

- 1. User logs in via SSO. Lands on Query Home with a search bar and recent query history.
- 2. User types: 'What are the most recent internal findings on compound X efficacy in Phase II trials?'
- 3. System retrieves top-N relevant chunks from vector DB. Retrieval layer detects semantic conflict between 3 chunks.
- 4. Conflict alert renders at the top of the answer: 'Evidence conflict detected across 3 studies. Study A (2019) reports 42% efficacy. Study C (2023) reports 61% efficacy under revised dosing protocol. Most recent evidence supports 61%. [Link to Study C, p.8]'
- 5. Below the conflict alert: synthesized answer with full inline citations [Study A, p.4], [Study B, p.12], [Study C, p.8].
- 6. Source Preview Panel opens — user clicks 'Expand Conflict Detail' to see the three conflicting passages side-by-side.
- 7. Confidence score: High. User is confident the conflict has been fully surfaced. Copies the conflict summary + citations into their research brief.
- 8. User follow-up: 'What explains the difference between the 2019 and 2023 findings?' — system retrieves methodology sections from both studies and synthesizes the delta.

Flow: Legal Counsel — Compliance Document Query

- 1. Legal user logs in. Sees 'Legal Mode' badge — audit logging confirmed active.
- 2. User queries: 'Are there any internal documents discussing adverse event disclosure obligations for compound Y?'
- 3. Answer returned with citations. Confidence: Medium — yellow banner shown.
- 4. User reviews sources. Clicks 'Flag for Review' on two specific source documents. Flagged to admin log.
- 5. User clicks 'Export Answer as PDF' — receives formatted report with citations, source metadata, and timestamp.
- 6. Report filed as supporting document in legal brief.

6.2 UI Design Principles

- Clarity over cleverness — the interface should feel like a search engine, not a chatbot.
- Conflicts are never hidden — contradiction alerts are top-of-answer, not buried in footnotes. If the evidence disagrees, the user sees that first.
- Recency is always visible — every source displays its date. Users should never have to wonder how old the evidence is.
- Trust through full traceability — confidence scores, source links, and the full trace view are always accessible. Nothing about how an answer was constructed is opaque.
- Minimal friction — zero-config for end users. No training, no Boolean syntax, no manual filters required to get a useful answer.
- Legal mode is visually distinct — different UI chrome so Legal users always know audit logging is active.

7. Technical Requirements

7.1 Architecture Overview

The SSE is built on an enhanced Retrieval-Augmented Generation (RAG) architecture with three layers beyond standard RAG: (1) recency-weighted retrieval scoring, (2) an inline contradiction detection pass over retrieved chunks, and (3) a full traceability layer that exposes every step of the evidence chain to the user. The system does NOT fine-tune a base model — all outputs are grounded in retrieved document context.

Component	Specification
Embedding Model	OpenAI text-embedding-3-large or equivalent (1536 dimensions). Must support batch embedding for ingestion pipeline.
Vector Database	Pinecone (managed) or pgvector (self-hosted). Metadata fields required: source_file, page_number, document_date, department, doc_type, chunk_index.
Retrieval Scoring	Hybrid score: (cosine_similarity * 0.75) + (recency_score * 0.25). Recency score derived from linear decay function over configurable time window (default 5 years). Weights configurable by admin.
Contradiction Detection	After retrieval of top-N chunks, a secondary LLM pass (lightweight, fast model) compares claim pairs for semantic opposition. Outputs: conflict_detected (bool), conflicting_chunk_ids[], conflict_summary, most_recent_chunk_id. Added latency budget: < 2 seconds.
LLM (Generation)	GPT-4o or Claude Sonnet via API. System prompt enforces: (a) citation-only generation, (b) surface contradiction alerts before synthesis, (c) label most-recent evidence explicitly.
Chunking Strategy	512-token chunks, 50-token overlap, paragraph-boundary aware. Tables and headers preserved as discrete chunks. Document date extracted and attached to all chunks from the same document.
Traceability Store	Each query response is stored with: query_id, retrieved_chunk_ids[], similarity_scores[], contradiction_result, llm_input_context, final_answer. Powers the Full Trace View UI.
Document Parser	Unstructured.io or LlamaParse for PDF/DOCX. OCR fallback for scanned documents. Date extraction from document headers, footers, and metadata fields.
Backend	Python / FastAPI. Async query handling. Horizontal scaling via container orchestration (Kubernetes).
Frontend	React + TypeScript. SSO integration (Okta or Azure AD). WCAG 2.1 AA accessibility compliant.
Infrastructure	AWS or Azure (client preference). All data encrypted at rest (AES-256) and in transit (TLS 1.3).
Latency Target	P95 query-to-answer latency < 12 seconds (including contradiction detection) under load of 50 concurrent users.
Uptime SLA	99.5% availability during business hours (7am–9pm ET, M–F).

7.2 Security & Compliance

- All document data remains within MediGen's approved cloud environment. No document content is sent to external APIs without explicit data processing agreement.
- LLM API calls transmit only retrieved text chunks as context — never full documents or PII.
- Role-based access control enforced at the API layer — Legal audit logs are not accessible via Research endpoints.
- HIPAA and 21 CFR Part 11 compliance requirements to be validated with MediGen's compliance team before Phase 1 data ingestion.
- Penetration testing to be completed before pilot launch (Week 8).

8. Dependencies, Constraints & Blockers

8.1 External Dependencies

Dependency	Detail & Risk
LLM API Provider	GPT-4o or Claude via API. Dependency on third-party availability and pricing. Mitigate: evaluate two providers, implement fallback routing.
SSO / Identity Provider	Okta or Azure AD integration required before any user can access the system. Must be provisioned by MediGen IT in Week 1.
Document Store Access	Engineering needs read access to MediGen's full 50,000-document repository by Day 3. Delayed access delays the entire ingestion timeline.
Data Processing Agreement (DPA)	DPA with LLM API provider must be signed before any MediGen document content is processed. Legal review required.
MediGen IT Sign-Off	Cloud infrastructure approval and network security review required before deployment. Target: Week 2.

8.2 Constraints

- Budget: MVP must be delivered within a fixed 3-month engagement. No scope expansion without written change order.
- Data residency: All MediGen data must remain in US-based cloud regions.
- No PII in corpus: Ingestion pipeline must flag and exclude any documents containing personally identifiable information pending review.
- Accessibility: Frontend must meet WCAG 2.1 AA standards before pilot launch.

8.3 Potential Blockers

Blocker	Mitigation
Document repository access delayed	Escalate to MediGen project sponsor on Day 1. No ingestion = no MVP. This is the critical path item.
LLM hallucination rate exceeds 1%	Tighten system prompt constraints; reduce chunk retrieval window; add mandatory citation validation layer before answer is surfaced.
DPA / legal approval takes > 2 weeks	Stand up ingestion pipeline in parallel using synthetic test data. Real corpus ingestion begins upon DPA execution.
SSO integration blocked by IT	Provide temporary username/password auth for pilot only. SSO required before production rollout.
Scanned PDFs with poor OCR quality	Flag problematic documents in Week 1 audit. Prioritize clean digital docs for MVP; OCR remediation in Phase 2.

9. Open Questions

Question	Owner & Target Resolution
What document metadata taxonomy is currently in use (if any) across the 50,000 docs?	MediGen IT — answer needed by Week 1 to design ingestion schema.
Does MediGen require on-premises or VPC-isolated LLM inference (vs. managed API)?	MediGen CISO — determines architecture path. Must resolve before Week 2.
What is MediGen's definition of a 'contradiction' — should methodological differences count, or only direct factual conflicts?	MediGen Research Lead + Tenex PM — calibrates contradiction detection sensitivity. Required before F-05 implementation begins.
What is the acceptable hallucination threshold for Legal use cases specifically?	MediGen Legal + Tenex PM — determines confidence threshold calibration.
When a contradiction is detected between an old internal study and a newer one, should the system recommend deprecating the older document?	MediGen Research Lead — determines whether contradiction detection feeds into corpus governance workflow.
Are any documents subject to external NDA or licensing restrictions that prohibit indexing?	MediGen Legal — must be resolved before ingestion begins.
Should the pilot include a feedback loop where users can rate answer quality and flag false-positive contradiction alerts?	Tenex PM — recommended; pending MediGen product team alignment.
What is MediGen's preferred export format for Legal audit reports — PDF only, or also DOCX?	MediGen Legal — low stakes; can be resolved in Week 3.

10. Risks & Mitigations

Risk	Likelihood	Mitigation
LLM generates an uncited or hallucinated answer accepted as fact by a legal user	Medium	Mandatory citation enforcement in system prompt; confidence scoring; human-review flag; Legal-tier users trained to treat Low-confidence answers as unverified.
Corpus contains outdated or conflicting documents causing inconsistent answers	High	This is now a feature, not a risk — contradiction detection surfaces these conflicts explicitly rather than silently averaging them into a misleading synthesis.
Contradiction detection produces false positives (flags genuine consensus as a conflict)	Medium	Tune semantic opposition threshold conservatively; label borderline cases as 'Possible conflict — verify manually'; collect false-positive feedback from pilot users to calibrate.
Document dates are missing or unreliable, breaking recency weighting	High	Pre-ingestion metadata audit in Week 1. For documents with no reliable date, fall back to file creation date and flag as 'Date unverified' in the source display.
User adoption stalls due to	Medium	Pilot with power users first; contradiction alerts and full

Risk	Likelihood	Mitigation
trust deficit in AI-generated answers		trace view directly address trust — they prove the system isn't a black box.
Ingestion pipeline fails on poorly formatted legacy documents	High	Pre-ingestion document quality audit in Week 1; error logging and quarantine for failed docs; human review queue.
Third-party LLM API experiences downtime during business hours	Low	Implement provider fallback (OpenAI primary, Anthropic secondary); cache answers for identical recent queries.
Scope creep from stakeholder feature requests during pilot	High	All new requests documented in a backlog and reviewed monthly. No mid-sprint scope changes. Formal change order process.

Appendix: Glossary

Term	Definition
RAG (Retrieval-Augmented Generation)	An AI architecture that retrieves relevant documents from a knowledge base and provides them as context to an LLM, grounding outputs in real sources rather than parametric memory.
Vector Database	A database optimized for storing and querying high-dimensional embeddings. Enables semantic similarity search across document chunks.
Embedding	A numerical representation of text that captures semantic meaning. Similar text produces similar embeddings, enabling relevance-based retrieval.
Chunking	The process of splitting documents into smaller segments for embedding. Chunk size and overlap affect retrieval quality.
Cosine Similarity	A metric for measuring semantic similarity between two embeddings. Used to rank retrieved document chunks by relevance.
Confidence Score	An SSE-specific score derived from the cosine similarity of top retrieved chunks, indicating how well the corpus supports the generated answer.
Audit Log	An immutable, timestamped record of queries, answers, and source citations — required for Legal-tier compliance.
MVP (Minimum Viable Product)	The smallest feature set that delivers measurable value to pilot users and validates the core product hypothesis.

— End of Document —